PARAMETRIZING A MODEL OF EFFICIENT CRYSTAL PRODUCTION

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Abstract. Surely this cannot be that difficult.

1. Introduction

A plethora of quantities arise in this work, so we endeavour to consistently indicate their nature at a glance. All constants $\|\cdot\|, \langle\cdot\rangle, |\cdot| \in \overline{\mathbb{R}^+}$ are non-negative, including scale $\|\cdot\|$ and stoichiometry $\langle\cdot\rangle$ (to avoid confusion, please note that there are no norms or inner products in this work). Any other constant $\cdot$ be written as $|\cdot|$. All italic quantities are dimensionless, to ease the unavoidable chore of tracking units. Ions are written without charge superscripts (e.g. OH means OH^-) to lighten notation, as this is a work of mathematics and not chemistry. Undefined notation adheres to the Digital Library of Mathematical Functions [1].

Imagine a complexing agent A interacting with cations $\mathrm{C}_I \in \left\{\mathrm{C}_1, \ldots, \mathrm{C}_{\|I\|}\right\}$ in a reactor to form an ionic complex

$$\mathrm{P} \rightleftharpoons (\mathrm{C}_1)_{\langle 1\rangle}(\mathrm{C}_2)_{\langle 2\rangle}\cdots(\mathrm{C}_{\|I\|})_{\langle\|I\|\rangle}(\mathrm{OH})_{\langle OH\rangle} \quad \text{in stoichiometric ratios} \quad \sum_{I=1}^{\|I\|}\langle I\rangle = 1$$

Let $|\mathrm{K_P}|$ denote the association (equilibrium) constant of complex P, which crystallizes when its concentration exceeds its equilibrium solubility $|\mathrm{K_S}|$. Crystallization is indicated by the supersaturation (ratio) exceeding unity

$$S^{\|I\|} := |\mathrm{K_P}|\,|\mathrm{K_S}|^{-1}\,[\mathrm{OH}]^{\langle OH\rangle}\prod_{I=1}^{\|I\|}[\mathrm{C}_I]^{\langle I\rangle} > 1 \tag{1.1}$$

where $[\cdot]$ denotes the free ion concentration, as distinct from the total species concentration $\Sigma[\cdot]$ across all ions.

Crystals of P are manufactured from a steady supply of stoichiometric amounts of C_I, A and other ingredients. These flow into a cation-free initial volume $\|\mathrm{V}\|\,(\mathrm{dm}^3)$ of complexing agent at a specially chosen initial concentration $\|\mathrm{A}\|\,(\mathrm{mol\,dm}^{-3})$. Defining

$$\begin{aligned}
\text{supply rate:}\quad & |\mathrm{F}| && (\mathrm{dm^3h^{-1}})\\
\text{complexing agent supply fraction:}\quad & |F_\mathrm{A}| < 1 &&\\
\text{complexing agent supply concentration:}\quad & \|\,\Sigma[\mathrm{A}]\| && (\mathrm{mol\,dm^{-3}})\\
\text{all cation supply fraction:}\quad & |F_\mathrm{C}| \le 1 - |F_\mathrm{A}| &&\\
\text{(individual) cation supply concentration:}\quad & \|\,\Sigma[\mathrm{C}_I]\| = \langle I\rangle\,\|\,\Sigma[\mathrm{C}]\| && (\mathrm{mol\,dm^{-3}})\\
\text{(variable) crystal yield:}\quad & \mathrm{P} && (\mathrm{mol})\\
\text{pH:}\quad & |H| &&
\end{aligned} \tag{1.2}$$

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Key words and phrases. Asymptotics.

This work was funded by Solomon f. Brown.

the manufacturing process amounts to

$$\begin{aligned}
\mathrm{V} &= \|\mathrm{V}\| + |\mathrm{F}|\,\mathrm{t} && (\mathrm{dm}^3)\\
\partial_{\mathrm{t}}\left(\mathrm{V}\,\Sigma[\mathrm{A}]\right) &= |\mathrm{F}|\,|F_{\mathrm{A}}|\,\|\,\Sigma[\mathrm{A}]\| && (\mathrm{mol\,h}^{-1})\\
\partial_{\mathrm{t}}\left(\mathrm{V}\,\Sigma[\mathrm{C}_I]\right) &= |\mathrm{F}|\,|F_{\mathrm{C}}|\,\langle I\rangle\,\|\,\Sigma[\mathrm{C}]\| - \langle I\rangle\partial_{\mathrm{t}}\mathrm{P} && (\mathrm{mol\,h}^{-1})\\
\Sigma[\mathrm{A}]_{\mathrm{t}=0} &= \|\mathrm{A}\| := |F_{\mathrm{A}}|\,\|\,\Sigma[\mathrm{A}]\| && (\mathrm{mol\,dm}^{-3})\\
\Sigma[\mathrm{C}_I]_{\mathrm{t}=0} &= 0 && (\mathrm{mol\,dm}^{-3})\\
\Sigma[\mathrm{P}]_{\mathrm{t}=0} &= 0 && (\mathrm{mol})
\end{aligned} \tag{1.3}$$

The specially chosen initial concentration is $\|\mathrm{A}\| := |F_{\mathrm{A}}|\,\|\,\Sigma[\mathrm{A}]\|\,(\mathrm{mol\,dm}^{-3})$ to keep $\Sigma[\mathrm{A}]$ constant. The all cation supply concentration is $\|\,\Sigma[\mathrm{C}]\| = \sum_{I=1}^{\|I\|}\langle I\rangle\,\|\,\Sigma[\mathrm{C}]\|\,(\mathrm{mol\,dm}^{-3})$, consisting of individual cations in stoichiometric ratios $\langle I\rangle$. The analysis which follows will admit changes of detail (changing ratios, outflow), but such considerations are best deferred: for clarity's sake alone we rigidly specify the system, relaxing unnecessary constraints only at the conclusion.

Crystal manufacture is modelled in three Sections: Supply, Saturation, and Crystallization.

2. Supply

Presumably cations are supplied in stoichiometry to avoid material waste. The resulting theoretical economy is delightful: every cation flows and precipitates identically. Non-dimensionalizing as

$$\begin{aligned}
t &:= \|\mathrm{V}\|^{-1}\,|\mathrm{F}|\,\mathrm{t}\\
V(t) &:= \|\mathrm{V}\|^{-1}\,\mathrm{V}\\
\Sigma A(t) &:= \|\mathrm{A}\|^{-1}\,\Sigma[\mathrm{A}]\\
\|\mathrm{C}\| &:= |F_{\mathrm{C}}|\,\|\,\Sigma[\mathrm{C}]\| && (\mathrm{mol\,dm}^{-3})\\
\Sigma C(t) &:= \langle I\rangle^{-1}\|\mathrm{C}\|^{-1}\,\Sigma[\mathrm{C}_I]\\
P(t) &:= \|\mathrm{C}\|^{-1}\,\|\mathrm{V}\|^{-1}\,\mathrm{P}
\end{aligned} \tag{2.1}$$

the dimensionless total species concentration $\Sigma C_I := \Sigma C$ is the same for all cations. Substituting Equation (2.1) in Equation (1.3) confirms the dimensionless supply solutions

$$\begin{aligned}
V(t) &= 1 + t\\
\Sigma A(t) &= 1\\
\Sigma C(t) &= \frac{t - P(t)}{1 + t}
\end{aligned} \tag{2.2}$$

In practice, mixing may be imperfect, delaying the spread of cations through the reactor volume. The spread is naturally monotonic, and we shall assume that is also linear and the same for all cations. Accordingly, the fraction of the reactor volume actually containing cations at time t is defined as

$$U(t) := \min\left(1, \frac{\|U\|\,t}{1 + t}\right) \quad \text{where} \quad \|U\| \geq |F_{\mathrm{C}}| \tag{2.3}$$

It is assumed that all other ingredients, including complexing agent A, mix perfectly and pervasively.

3. Saturation

Ionic complex formation is reversibly instantaneous. The aqueous complexing agent A complexes with protons H and cations C_I to form

$$\begin{aligned}
(\mathrm{A})(\mathrm{H}) &\quad \text{with equilibrium constant } \mathrm{K}_{\mathrm{H}}.\\
(\mathrm{C}_I)(\mathrm{A})_j &\quad \text{with equilibrium constant } \mathrm{K}_{Ij}.
\end{aligned} \tag{3.1}$$

The dimensional mass balances for A and C therefore take the familiar form

$$\begin{aligned}
\Sigma[\mathrm{A}] &= [\mathrm{A}] + \mathrm{K}_{\mathrm{H}}[\mathrm{A}][\mathrm{H}] + \sum_{I=1}^{\|I\|}[\mathrm{C}_I]\sum_{j=1} j\mathrm{K}_{Ij}[\mathrm{A}]^j && (\mathrm{mol\,dm}^{-3})\\
\Sigma[\mathrm{C}_I] &= [\mathrm{C}_I] + [\mathrm{C}_I]\sum_{j=1}\mathrm{K}_{Ij}[\mathrm{A}]^j && (\mathrm{mol\,dm}^{-3})
\end{aligned} \tag{3.2}$$

The non-dimensionalization here is as simple as can be: free ion concentration is tracked as a proportion of total concentration.

$$\begin{aligned}
\tau &:= \sum_{I=1}^{\|I\|} \langle I \rangle \ln[\mathrm{C}_I] - \sum_{I=1}^{\|I\|} \langle I \rangle \ln\, \Sigma[\mathrm{C}_I] \\
\eta &:= \log \mathrm{K_H} + 14 - |H| \\
a(\tau) &:= (1 + 10^{\eta})[\mathrm{A}]\, \Sigma[\mathrm{A}]^{-1} \\
K_{Ij} &:= \mathrm{K}_{Ij}(1 + 10^{\eta})^{-j}\, \|\mathrm{A}\|^{j} = \mathrm{K}_{Ij}[\mathrm{A}]^{j} a(\tau)^{-j}
\end{aligned} \tag{3.3}$$

This simple formulation has a powerful consequence: mass balance equations adopt the form of mathematical derivatives with respect to supply (i.e. total concentration). For each cation $\mathrm{C} \in \{\mathrm{Z}, \mathrm{Y}, \ldots\}$ define the polynomial

$$\begin{aligned}
\Xi_I(a(\tau)) &:= 1 + \sum_{j=1} K_{Ij} a(\tau)^j \\
\Xi(a(\tau)) &:= \prod_{I=1}^{\|I\|} \Xi_I(a(\tau))^{\langle I \rangle}
\end{aligned} \tag{3.4}$$

Dimensionless mass balances translate Equation (3.2) into

$$\begin{aligned}
\tau &= -\ln \Xi(a(\tau)) \\
(1 - a(\tau))\, \partial_\tau \ln a(\tau) &= -\|\mathrm{C}\|\, \|\mathrm{A}\|^{-1}\, \Sigma C(t)
\end{aligned} \tag{3.5}$$

primes indicating differentiation according to Lagrange/Euler notation.

Saturation is indicated by

$$\|\mathrm{S}\| s(t) > 1 \tag{3.6}$$

where

$$\begin{aligned}
s(t) &:= \prod_{\mathrm{C}} \left(c(t)\, \Sigma c(t) U(t)^{-1} \right)^{C/S} = \|\mathrm{S}\|^{-1}\, \mathrm{S} \\
\|\mathrm{S}\|^{S} &:= \mathrm{K_P K_S^{-1}} 10^{(h-14)H}\, \|\oplus\| \prod_{\mathrm{C}} C^{C}
\end{aligned} \tag{3.7}$$

Note that saturation is local to the volume fraction $U(t)$ containing cations. The total cation concentration in this region is conveniently expressed as

$$\begin{aligned}
\Sigma c(t) U(t)^{-1} \; &= p(t)\, \|U\|^{-1} \max\left(1, \frac{\|U\|\, t}{1+t}\right) \\
\text{where} \quad p(t) &:= 1 - t^{-1} P(t)
\end{aligned} \tag{3.8}$$

Let us expand about $t = \overline{t}$ using

$$\begin{aligned}
\overline{a} &:= a(\overline{t}) \\
\overline{p} &:= p(\overline{t}) \\
\overline{u} &:= \Sigma c(\overline{t}) = \overline{p}\overline{t}(1+\overline{t})^{-1} \\
\overline{U} &:= U(\overline{t}) = \min\left(1, \|U\|\, \overline{p}^{-1} \overline{u}\right) \\
\overline{s} &:= s(\overline{t}) \\
\|E\| &:= \|\oplus\|\, \|\mathrm{A}\|^{-1} \\
E(\#) &:= \|E\| \sum_{\mathrm{C}} C\, \ln \Xi_{\mathrm{C}}(\#) \\
\overline{E'} &:= \overline{u}\, \overline{a} E'(\overline{a}) \quad ; \quad \overline{E''} := \overline{u}\, \overline{a}^2 E''(\overline{a}) \quad ; \text{etc.} \\
\tilde{u}(t) &:= \Sigma c(t) - \overline{u} \\
\|\tilde{u}\| &:= \overline{E'}^{-1} \left(1 + \overline{E''}\right)^2 \overline{u} \\
u(t) &:= \|\tilde{u}\|^{-1}\, \tilde{u}(t)
\end{aligned} \tag{3.9}$$

Equation (3.5) becomes

$$\begin{aligned}
a(t)^{-1} &= \overline{a}^{-1} + \tilde{u}(t) E'(a(t)) + \overline{u}\, (E'(a(t)) - E'(\overline{a})) \\
S \ln s(t) &= S \ln \overline{s} + \ln \frac{\tilde{u}(t) + \overline{u}}{\overline{u}} - \ln \frac{U(t)}{\overline{U}} - \frac{E(a(t)) - E(\overline{a})}{\|E\|}
\end{aligned} \tag{3.10}$$

which has the asymptotic solution

$$\frac{a(t)}{\overline{a}} = 1 - u(t)\left(1+\overline{E''}\right)\left(1 - u(t)\left(\frac{\overline{E''}}{\overline{E'}}\left(1+\overline{E''}\right) + 1 - \frac{\overline{E'''}}{2}\right) + O\left(u(t)^2\right)\right)$$

$$S\ln\frac{s(t)}{\overline{s}} = \ln\frac{p(t)}{\overline{p}} + \left(\ln\frac{\|U\|\,t}{1+t}\right)^+ - \left(\ln\frac{\|U\|\,\overline{t}}{1+\overline{t}}\right)^+ \tag{3.11}$$
$$+ u(t)\frac{\left(1+\overline{E''}\right)^3}{\|\tilde{u}\|\,\|E\|}\left(1 - u(t)\left(\frac{3\overline{E''}}{2\overline{E'}}\left(1+\overline{E''}\right) + 1 - \frac{\overline{E'''}}{2}\right) + O\left(u(t)^2\right)\right)$$

The point of expansion $\overline{t}$ is chosen to achieve a saturation ratio $\overline{\mathrm{S}}$ by (numerically) solving

$$S\ln\|\mathrm{S}\| + S\ln\overline{s} =: S\ln\overline{\mathrm{S}} = S\ln\|\mathrm{S}\| + \ln\overline{p} - \ln\|U\| + \left(\ln\frac{\|U\|\,\overline{t}}{1+\overline{t}}\right)^+ - \|E\|^{-1}E(\overline{a}) \tag{3.12}$$
$$\overline{a}^{-1} = a(0)^{-1} + \overline{u}E'(\overline{a})$$

The positive part $(\#)^+$ in these equations is 0 unless $\overline{U} = 1$.

References

1. *NIST Digital Library of Mathematical Functions*, `https://dlmf.nist.gov/`, Release 1.2.6 of 2026-03-15, F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.